

As you have probably figured out by now, Augmented Reality depends a great deal on the hardware. In fact you may even say it's vital to its existence. In this chapter we will introduce you to the tools of the trade, the machinery which together will shape the way we will perceive our world in the future.

The basic kit

Although augmented reality has a lot of uses in various fields, they all have few similarities that cannot be ignored. They all need some sort of input which needs to be augmented. This input is fetched from the camera, most of the times. Next comes the part about deciding what graphical information needs to be shown on screen; depending on the application, the magnitude of annotations vary. However, in all cases some sensing devices are required to fetch information about what needs to be displayed. Depending on the type of interaction that the AR application needs to do with the user, it may need other input devices (for e.g. a tourism AR browser might feature a touch-to-learn feature). Next comes the dough of AR: a processor which combines the visual data being fed into the system with the processed changes that need to be displayed. Since we are in need of displaying the augmentation, AR also needs a display in most cases.

One size does not fit all

Depending on what the application is, in addition to the common hardware we mentioned above, it might need some more. For instance, the use of AR in tourism would need inputs from a GPS unit, an accelerometer and even a digital compass. On the other hand, usage of AR in military might require additional tagging and real world path highlighting. It might need information from custom devices, the local map and a few other information sources to assist the soldier. A virtual shopping trial room would work well enough with a webcam. In this chapter we take a look at the Hardware involved in AR, starting with display.

Displays

There are numerous ways to show the augmentation in AR systems and it is affected by the usage scope of the particular AR application. As per use-case the screen will differ. Let's find out how.

Head-mounted display (HMD)

Head-mounted displays are the type of displays which can be mounted